

Human Embryonic Stem Cells Suffer from Centrosomal Amplification

ZUZANA HOLUBCOVÁ,^a PAVEL MATULA,^b MIROSLAVA SEDLÁČKOVÁ,^c VLADIMÍR VINARSKÝ,^{a,d} DÁŠA DOLEŽALOVÁ,^{a,d} TOMÁŠ BÁRTA,^{a,d} PETR DVOŘÁK,^{a,d} ALEŠ HAMPL^{c,d}

^aDepartment of Biology, Faculty of Medicine, Masaryk University, Brno, Czech Republic; ^dDepartment of Molecular Embryology, Institute of Experimental Medicine, v.v.i., Academy of Sciences of the Czech Republic, Brno, Czech Republic; ^bCentre for Biomedical Image Analysis, Faculty of Informatics, Masaryk University, Brno, Czech Republic; ^cDepartment of Histology and Embryology, Faculty of Medicine, Masaryk University, Brno, Czech Republic

Key Words. Human embryonic stem cells • Centrosome • Chromosome • Genomic instability • CDK • Aurora A

ABSTRACT

Propagation of human embryonic stem cells (hESCs) in culture tends to alter karyotype, potentially limiting the prospective use of these cells in patients. The chromosomal instability of some malignancies is considered to be driven, at least in part, by centrosomal overamplification, perturbing balanced chromosome segregation. Here, we report, for the first time, that very high percentage of cultured hESCs has supernumerary centrosomes during mitosis. Supernumerary centrosomes were strictly associated with an undifferentiated hESC state and progressively disappeared on prolonged propagation in culture. Improved attachment to culture substratum and inhibition of CDK2 and Aurora A (key regulators of centrosomal metabolism)

diminished the frequency of multicentrosomal mitoses. Thus, both attenuated cell attachment and deregulation of machinery controlling centrosome number contribute to centrosomal overamplification in hESCs. Linking the excessive number of centrosomes in mitoses to the ploidy indicated that both overduplication within a single cell cycle and mitotic failure contributed to generation of numerical centrosomal abnormalities in hESCs. Collectively, our data indicate that supernumerary centrosomes are a significant risk factor for chromosome instability in cultured hESCs and should be evaluated when new culture conditions are being implemented. *STEM CELLS* 2011;29:46–56

Disclosure of potential conflicts of interest is found at the end of this article.

INTRODUCTION

One hurdle that may limit future medical exploitation of pluripotent cells such as human embryonic stem cells (hESCs) is the instability of their genetic complement. The consequences of this instability [1] and its underlying mechanisms are largely unknown. The dynamics of the development of chromatin changes in cultured hESCs suggest that these changes participate in stepwise adaptation of embryo-derived cells to suboptimal in vitro conditions. Nevertheless, the mechanisms by which modifications to the genome and epigenome of hESCs are generated and stabilized in the cell population remain enigmatic. In principle, they may be produced by intrinsic and extrinsic factors predisposing cells to genetic mutation, less effective surveillance mechanisms, or combination of both.

Among the genetic alterations that have been clearly linked to long-term propagation of hESCs, chromosomal abnormalities (aneuploidies and translocations) have been studied the most and are the best characterized [2–4]. Such

genome alterations are proposed to be fuelled by mitotic chromosomal instability, that is, the inability to faithfully segregate a correct chromosome complement to two daughter cells during cell division [5]. Notably, chromosomal instability, which is a typical feature of many human cancers, is frequently associated with abnormalities in the number and structure of centrosomes, the principle microtubule-organizing centers (MTOCs) in cells [6–8].

The mammalian centrosome is a tiny unbounded organelle containing two barrel-shaped centrioles embedded in an amorphous mass of pericentriolar material (PCM) [9]. As schematized in Figure 1A, the life cycle of a centrosome is coordinated with that of DNA, with the centrosome also being semiconservatively duplicated just once per cell cycle (reviewed in [10, 11]). Being the major MTOC, centrosomes influence all microtubule-dependent processes including organelle transport, cell shape, cell polarity, and chromosome segregation [12, 13]. Although a causal relationship between centrosomal abnormalities and tumorigenesis still awaits to be clearly established, a direct mechanistic link between extra

Author contributions: Z.H.: conception and design, collection and assembly of data, data analysis and interpretation, manuscript writing; P.M.: collection and assembly of data, data analysis and interpretation; M.S.: collection and assembly of data, data analysis and interpretation; V.V.: data analysis and interpretation; D.D.: data analysis and interpretation; T.B.: data analysis and interpretation; P.D.: data analysis and interpretation; A.H.: conception and design, data analysis and interpretation, manuscript writing, final approval of manuscript.

Correspondence: Aleš Hampl, D.V.M., PhD., Department of Histology and Embryology, Faculty of Medicine, Masaryk University, Kamenice 5, Brno 62500, Czech Republic. Telephone: 420-549-49-3514; Fax: 420-549-49-8001; e-mail: ahampl@med.muni.cz Received August 2, 2010; accepted for publication September 19, 2010; first published online in *STEM CELLS EXPRESS* October 19, 2010. © AlphaMed Press 1066-5099/2009/\$30.00/0 doi: 10.1002/stem.549

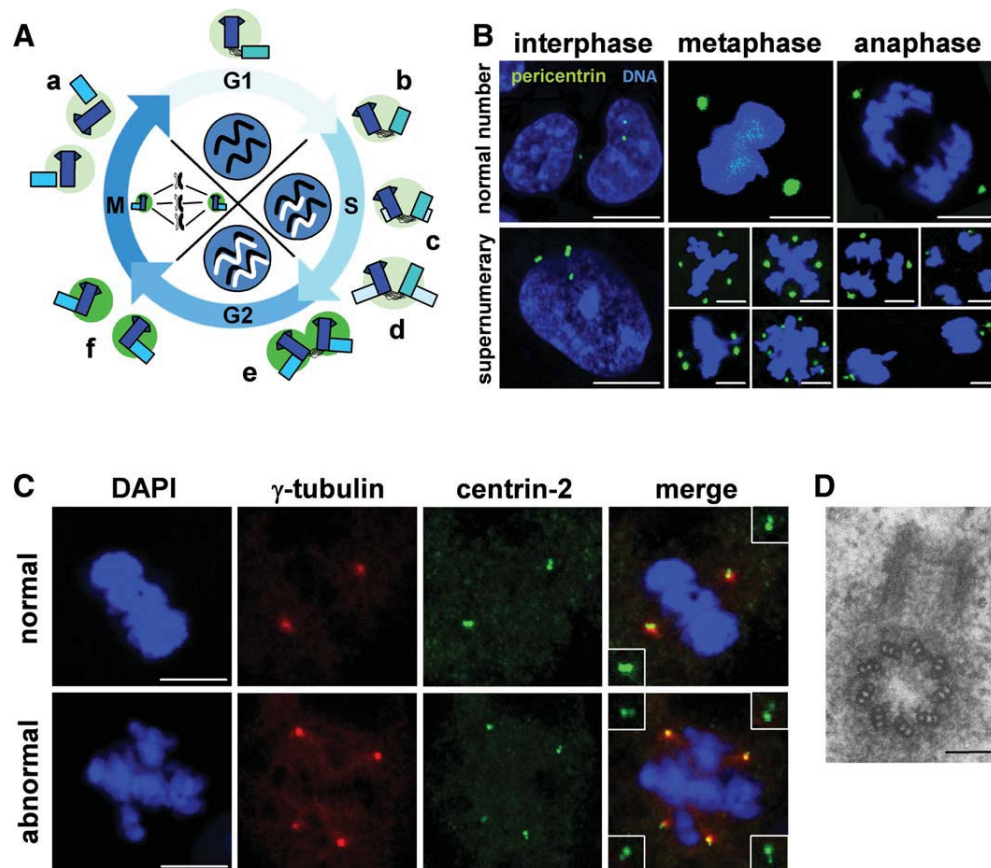


Figure 1. Numerical abnormalities in centrosomes in human embryonic stem cells (hESCs) at different stages of the cell cycle. (A): Schematic of the key steps (a–f) in centrosome metabolism, as related to the cell cycle phases. (a): Centrioles disengage, licensing centrosome duplication. (b): Centrioles then split slightly and lose their mutually orthogonal position. (c): Procentrioles form on each pre-existing centriole and (d) elongate. (e): Centrosomes accumulate additional pericentriolar material and thus mature. (f): Fully mature centrosomes separate and serve as microtubule organizing centers. (B): Examples of hESCs with normal and supernumerary centrosomes, as observed at interphase, metaphase, and anaphase. Chromatin was stained with DAPI (blue) and centrosomes were labeled with anti-pericentrin antibody (green). Scale bar = 5 μ m. (C): Two centrioles are invariably seen in normal and supernumerary centrosomes. Chromatin was stained with 4,6'-diamidino-2-phenylindole (DAPI), (blue). Centrosomes were labeled with anti- γ -tubulin antibody (red) and individual centrioles with anti-centrin antibody (green). Scale bar = 5 μ m. (D): Normal ultrastructure of centrosomes making up hESC centrosomes, as visualized by TEM. Scale bar = 0.1 μ m. Abbreviation: DAPI, 4,6'-diamidino-2-phenylindole.

centrosomes and chromosomal instability has been recently demonstrated experimentally [14, 15].

Given the critical importance of sustained genome integrity in hESCs, we investigated centrosome metabolism in these cells. We show, in a large panel of independent hESC lines propagated in two different laboratories, that pristine hESCs develop numerical, but not structural, abnormalities of centrosomes. We also identify several factors that contribute to this phenomenon and at least two different underlying mechanisms.

MATERIALS AND METHODS

Cell Culture

Undifferentiated hESCs were maintained on mitotically inactivated mouse embryonic fibroblasts (mEFs) or inactivated human foreskin fibroblasts (hFFs) in culture media consisting of Dulbecco's modified Eagle's medium (DMEM)/F12 supplemented with 15% knockout serum replacement, L-glutamine, minimum essential medium (MEM) nonessential amino acids, 0.5% penicillin-

streptomycin, (all from Invitrogen, Carlsbad, CA), β -mercaptoethanol (Sigma-Aldrich, St. Louis, MO), and 10 ng/ml fibroblast growth factor (FGF)-2 (PeproTech, Rocky Hill, NJ). The following lines of hESCs were grown on mEFs: CCTL6, CCTL8, CCTL10, CCTL12, CCTL13, and CCTL14 [16]. The following hESC lines were grown on hFFs and were obtained as fixed samples: HS181, HS207, HS306, HS401, HS420, and HUES9 (Dr. Outi Hovatta, Karolinska Institutet, Stockholm, Sweden).

hFFs and hESC-derived fibroblast-like cells (derived from the CCTL14 hESC line) were cultured in KnockOut DMEM supplemented with 10% fetal bovine serum (FBS), L-glutamine, MEM nonessential amino acids, 1% penicillin-streptomycin (all from Invitrogen), and β -mercaptoethanol (Sigma-Aldrich).

β 3Tu⁺/Pax6⁺ hESC-derived cells (derived from the CCTL14 hESC line) were cultured in DMEM/F12 supplemented with L-glutamine, MEM nonessential amino acids, N2 and B27 supplements, 0.5% penicillin-streptomycin, (all from Invitrogen), β -mercaptoethanol (Sigma-Aldrich), and 20 ng/ml FGF-2 (PeproTech).

Differentiation of hESCs

To induce differentiation of hESCs under adherent conditions, cells growing in standard hESC media were treated with 5 μ M

retinoic acid (RA; R2625, Sigma-Aldrich, St. Louis, MO, www.sigmaaldrich.com) for 3 days.

To induce differentiation of hESCs via embryoid body (EB) formation, colonies of undifferentiated hESCs were detached and placed onto nonadhesive dishes and cultured for 5 days in Knock-Out DMEM supplemented with 20% FBS, L-glutamine, MEM nonessential amino acids, 1% penicillin-streptomycin (all from Invitrogen, Carlsbad, CA, www.invitrogen.com), and β -mercaptoethanol (Sigma-Aldrich). Resulting EBs were then placed onto adhesive dishes and allowed to differentiate in the same media for another 10 days.

Cell Treatments

The Aurora A chemical inhibitor, MLN8054 [17] (Millennium Pharmaceuticals, Cambridge, MA, www.millennium.com; provided by Dr. Spiros Linardopoulos), and the CDK2 chemical inhibitors, roscovitine (R7772, Sigma-Aldrich) and olomoucine II [18] (provided by Dr. Miroslav Strnad, Palacký University Olomouc, Olomouc, Czech Republic), were prepared as 1,000 $\times$ stocks in DMSO and were added directly to the culture media. DMSO (0.1%) served as a vehicle control. Anti-LIBS-6 $\alpha_{\text{LIBS-6}}$ integrin receptor-activating antibody [19, 20] (provided by Dr. Mark Ginsberg, University of California, San Diego, CA) was prepared as a 2 mg/ml stock in phosphate-buffered saline (PBS; pH 7.4) and was directly added to the culture media at a final concentration of 10 $\mu\text{g/ml}$.

Cultureware and Coatings

The cell culture chambers/dishes that were used in culture surface modification experiments were the Lab-Tek II Chamber Slide System (154526, Nalge Nunc International Corp., Naperville, IL, www.nalgenunc.com) and polyethylene-based μ -dishes (81156, ibidi, Munich, Germany, www.ibidi.com). The following coating solutions were used: 0.1% gelatin (G1890, Sigma-Aldrich) in water or a mixed solution of 10 $\mu\text{g/ml}$ fibronectin (33016015, Invitrogen) and 10 $\mu\text{g/ml}$ laminin (L2020, Sigma-Aldrich) in PBS (pH 7.4). All coatings were carried out overnight at 37°C.

Indirect Immunofluorescence

For immunodetection of pericentrin and Oct3/4, cells were fixed in an 95% ethanol/1% acetic acid solution for 30 minutes on ice, transferred stepwise into PBS (pH 7.4), permeabilized for 15 minutes with 0.1% Triton-X, blocked for 1 hour with 1% bovine serum albumin in PBS (pH 7.4) containing 0.05% Tween-20, and incubated overnight at 4°C with primary antibody against pericentrin (ab4448, abcam, Cambridge, U.K.) and/or Oct3/4 (sc-5279, Santa Cruz Biotechnology, Santa Cruz, CA, www.scbt.com).

For immunodetection of centrin and γ -tubulin, cells were fixed in cold methanol:acetone (9:1) solution for 20 minutes at -20°C , transferred into PBS (pH 7.4), and further processed as mentioned earlier. The following primary antibodies were used: anti-centrin-2 (sc-27793-R, Santa Cruz Biotechnology) and/or anti- γ -tubulin (TU-31, provided by Dr. Pavel Dráber, Institute of Molecular Genetics, Academy of Sciences of the Czech Republic, Prague, Czech Republic).

For immunodetection of kinetochores, cells were fixed in 2% paraformaldehyde for 5 minutes at room temperature and in cold methanol for 5 minutes at -20°C . They were then washed for 5 minutes in MBST buffer containing 10 mM 3-(*N*-Morpholino)-propanesulfonic acid (pH 7.2), 150 mM NaCl, and 0.05% Tween-20. The cells were blocked with 5% FBS in MBST buffer and incubated with human anti-centromere autoantibody (CREST, 90C-CS-1058, Europa Bioproducts, Cambridge, U.K., www.europa-bioproducts.com) diluted in MBST buffer containing 2% FBS.

Final incubations with secondary antibodies were carried out for 1 hour at room temperature. Cell nuclei were counterstained with 4',6-diamidino-2-phenylindole (DAPI, Sigma-Aldrich), and cells were mounted in Mowiol containing 1,4-diazobicyclo[2.2.2.]

octane to prevent fading. Microscopy was performed using an Olympus FluoView 500 laser scanning microscope (Olympus C&S Ltd., Prague, Czech Republic, <http://www.olympus-global.com>).

Ploidy Determination

To analyze the relationship between centrosome counts and ploidy in individual cells, we simultaneously immunostained hESCs for pericentrin and kinetochores. The preparations were observed under an Olympus FluoView 500 laser scanning microscope, and identified multicentrosomal mitoses were Z-scanned (step 0.13 μm) using a $\times 100$ objective. Acquarium software [21] was used to determine the numbers of kinetochores. The plug-in "Find small objects (eMax)" feature was used to detect individual kinetochores (<http://cbia.fi.muni.cz/acquarium/>).

Electron Microscopy

Cells were fixed in 300 mM glutaraldehyde (G7651, Sigma-Aldrich) in 100 mM cacodylate buffer for 60 minutes at room temperature and postfixed with 40 mM osmium tetroxide (75630, Sigma-Aldrich) in 100 mM cacodylate buffer for 60 minutes at room temperature. Cells were then dehydrated in ethanol and embedded in situ in Durcupan ACM Fluka (44610, Sigma-Aldrich). Blocks containing hESC colonies were excised using a pad saw. Ultrathin sections (40 nm) were cut in parallel to the monolayer using a Leica Ultracut UCT ultramicrotome and stained with uranyl acetate and lead citrate. Sections were examined under a FEI Morgagni 268D electron microscope (FEI Company, Eindhoven, The Netherlands, www.fei.com).

Western Blot Analysis

hESC colonies were harvested by manual scraping, washed three times with PBS (pH 7.4) and lysed in 100 mM Tris-HCl (pH 6.8) containing 20% glycerol and 1% SDS. Protein concentrations were determined using the DC Protein Assay Kit (Bio-Rad, Hercules, CA, www.bio-rad.com). Lysates were supplemented with bromophenol blue (0.01%) and β -mercaptoethanol (1%). Equal amounts of total protein were separated by SDS-polyacrylamide gel electrophoresis (PAGE), electrotransferred onto polyvinylidene fluoride (PVDF) membranes (Millipore, Billerica, MA, www.millipore.com), immunodetected using appropriate antibodies, and visualized using ECL Plus reagent (GE Healthcare Life Sciences, Little Chalfont, U.K., www.gelifsciences.com) according to the manufacturer's instructions. Primary antibodies were as follows: rabbit anti-cyclin E1 (MS-870-P) and anti-cyclin A2 (MS-384-P1) polyclonal antibodies (Neomarkers, Fremont, CA, www.labvision.com), mouse anti-Aurora A monoclonal antibody (ab13824, Abcam), and mouse anti- α -tubulin monoclonal antibody (11-250-C100; Exbio, Vestec, Czech Republic, www.exbio.cz).

Immunoprecipitation and Kinase Assay

Cells were lysed in ice-cold extraction buffer (50 mM Tris-HCl, 150 mM NaCl, 0.5% Nonidet P-40, 1 mM EDTA, 8 mM β -glycerophosphate, 50 mM NaF, 1 mM phenylmethylsulfonyl fluoride, 1 mM tosylphenylalanine chloromethane, 1 mM dithiothreitol, 1 $\mu\text{g/ml}$ leupeptin, 1 $\mu\text{g/ml}$ aprotinin, and 10 $\mu\text{g/ml}$ soybean trypsin inhibitor) and the extracts were stored at -80°C until further use. Before use, extracts were thawed and cleared by centrifugation at 15,000g for 5 minutes at 4°C. Protein concentrations were determined using the DC Protein Assay Kit (Bio-Rad). Extracts were incubated for 1 hour on ice with rabbit polyclonal antibody against CDK2 (sc-163; Santa Cruz Biotechnology) to immunoprecipitate CDK2. Immunoprecipitates were collected on Protein G agarose beads (P7700, Sigma-Aldrich) by overnight rotation at 4°C. They were then washed three times with extraction buffer and twice with kinase assay buffer (50 mM HEPES [pH 7.5], 10 mM MnCl_2 , 8 mM β -glycerophosphate, and 1 mM dithiothreitol). Kinase reactions were carried out for 30 minutes

Table 1. Undifferentiated human embryonic stem cells

Cell line	Passage number	Mitoses multicentrosomal/total	Multicentrosomal mitoses percentage (%)
CCTL6	P26	18/88	20.45
CCTL8	P24	31/201	15.40
CCTL10	P14	61/260	23.46
CCTL12	P18	17/158	10.82
CCTL13	P18	10/68	14.70
CCTL14	P19	38/237	16.03
HS181	P25	18/108	16.60
HS420	P31	21/172	12.02
HS207	P27	10/61	14.75
HS306	P39	21/131	16.03
HS401	P23	15/146	10.27
HUES9	P27	57/544	10.47

at 37°C in a total volume of 25 μ l of kinase assay buffer supplemented with 100 μ g/ml histone H1 (Sigma-Aldrich) and 40 μ Ci/ml [32 P]ATP. Reactions were terminated by addition of 2 $\times$ Laemmli sample buffer and subjected to SDS-PAGE and autoradiography.

RESULTS

hESCs Develop Numerical Abnormalities in Their Centrosomes

Abnormalities in the mitotic spindle may lead to unequal distribution of a genetic material during cell division. One such abnormality is a multipolar spindle, which very often develops in cancer cells and is associated with the presence of supernumerary centrosomes [6–8]. In our initial search for potential contributors to chromosomal instability in hESCs, we have noticed that both interphasic and mitotic cells very often have inadequate numbers of centrosomes, as visualized by pericentrin staining (Fig. 1B). As pericentrin is just one component of PCM in this organelle, we confirmed the presence of supernumerary centrosomes by staining for another PCM protein, γ -tubulin (supporting information Fig. S1). In mitotic hESCs, multicentrosomal states (>2 centrosomes per cell) were almost invariably associated with major abnormalities in the chromosomal positioning (Fig. 1B), although clustering of supernumerary centrosomes resulting in a pseudobipolar spindle was also observed. The presence of multipolar spindles in hESCs with supernumerary centrosomes (supporting information Fig. S2) indicated that these extraneous organelles were fully functional as MTOCs. Next, centrosome structure was analyzed by staining for the centriole-specific protein centrin and by transmission electron microscopy (TEM). As shown in Figure 1C, staining with anti-centrin antibody revealed the presence of two centrioles in all normal and supernumerary centrosomes. Disruptions in centriole barrel structure, unincorporated microtubule complexes, centrioles of unusual length or mispositioning of centrioles can be found by TEM in cancer cells [22]. In our study, of the 130 centrosomes that were identified in TEM images, none showed any signs of such structural anomalies (Fig. 1D).

The Presence of Supernumerary Centrosomes Is Common to Many Independent hESC Lines

A variety of factors may contribute to the occurrence of supernumerary centrosomes, including hESC line genotype, technique of derivation, and culture routine. Therefore, we extended

Table 2. Differentiated cell types

Cell type	Mitoses multicentrosomal/total	Multicentrosomal mitoses percentage (%)
Human foreskin fibroblasts SCRC 1041	5/245	2.04%
hESC-derived fibroblast-like cells	1/37	2.70%
β 3Tu $^{+}$ /Pax6 $^{+}$ hESC-derived cells	5/106	4.71%

Abbreviation: hESC, human embryonic stem cell.

our analysis to 12 independent hESC lines that had different origins and had been cultured in different laboratories to assess the generality of this phenomenon. As summarized in Table 1, the percentage of multicentrosomal mitoses ranged from about 10% to 24% in low-passage hESCs (from P14 to P39). Thus, a high frequency of multicentrosomal mitoses is a typical feature of hESCs rather than a product of the specific genotype and/or actual culture conditions. Although supernumerary centrosomes are commonly found in many cancer cells, the frequency of such abnormalities in normal cells is an unexplored issue. Therefore, we also determined the frequency of multicentrosomal mitoses in three different types of non-hESC/nonmalignant cell types, that is, hFFs, hESC-derived fibroblast-like cells, and β 3Tu $^{+}$ /Pax6 $^{+}$ hESC-derived cells. As shown in Table 2, this frequency varied between about 2% and 5%. Thus, it was two-fold to fivefold lower than that in hESCs.

Numerical Abnormalities in Centrosomes Become Suppressed on Entry of hESCs into Differentiation Pathways or on Prolonged Propagation in Culture

The normal centrosomal status found here in fibroblast-like cells derived in vitro from hESCs (Table 1B) suggests that only the pristine, pluripotent state is associated with a high frequency of numerical centrosomal abnormalities. To explore this possibility more rigorously, we exposed hESCs to two differentiation-inducing cues. Specifically, we treated cells with RA or induced EB formation in non-adherent cultures. Importantly, both differentiation scenarios produced cells with low Oct3/4 expression and a frequency of multicentrosomal mitoses below 5% (Fig. 2).

Several studies have investigated the changes that hESCs in a nondifferentiated state undergo during prolonged propagation in culture and how these changes may underlie the culture-adaptation phenomenon [23]. Given that supernumerary centrosomes are likely to produce a major stress to hESCs and that adaptation of hESCs seems to increase their fitness, we reasoned that a link may exist between these two phenotypic entities. To address this hypothesis, we analyzed the frequency of multicentrosomal mitoses in hESC that had been cultured for varying numbers of passages. Although differences were detected between hESCs cultured on gelatinized glass and gelatinized polyethylene-based dishes (μ -dishes), the overall trend fit our expectations very well (Fig. 3A, 3B); that is, under both conditions, the frequency of multicentrosomal mitoses was highest in the earliest passages and then continuously decreased until reaching a minimum of about 5% in later passages.

Cues from the Culture Substratum Affect the Frequency of Numerical Centrosomal Abnormalities

As mentioned earlier, there was a noticeable variability in the frequencies of multicentrosomal mitoses found in hESCs, and this variability was associated with the type of material used

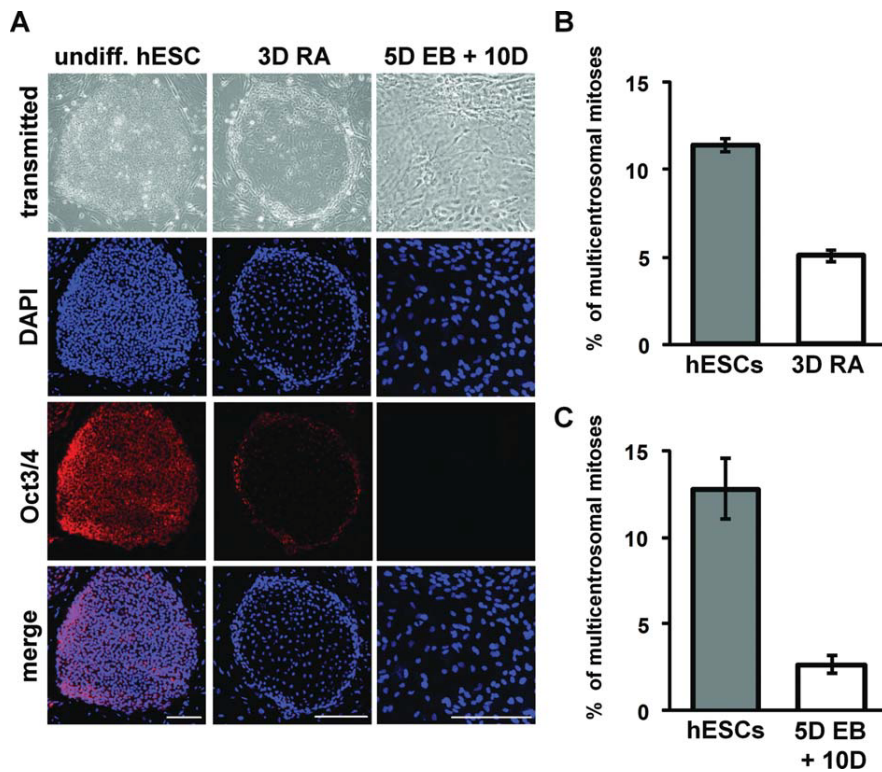


Figure 2. Numerical abnormalities in centrosomes become suppressed in differentiating/differentiated hESCs. (A): Differentiation of hESCs was induced (a) by exposure to RA for 3 days or (b) by formation of embryoid bodies (EBs) for 5 days followed by adherent growth for 10 days. Cellular morphology and expression of Oct3/4 (red) are shown to demonstrate undifferentiated and differentiated states. Chromatin stained with 4',6-diamidino-2-phenylindole (DAPI) is shown in blue. Scale bar = 100 μ m. (B): Decrease in the percentage of mitoses with supernumerary centrosomes in differentiated hESCs (by RA) compared with undifferentiated controls. (C): Decrease in the percentage of mitoses with supernumerary centrosomes in differentiated hESCs (by EB formation) compared with undifferentiated controls. Data are expressed as the mean $\pm$ SEM of three independent replicates. Abbreviations: DAPI, 4',6'-diamidino-2-phenylindole; EB, $\bullet\bullet\bullet$; hESC, human embryonic stem cell; RA, retinoic acid.

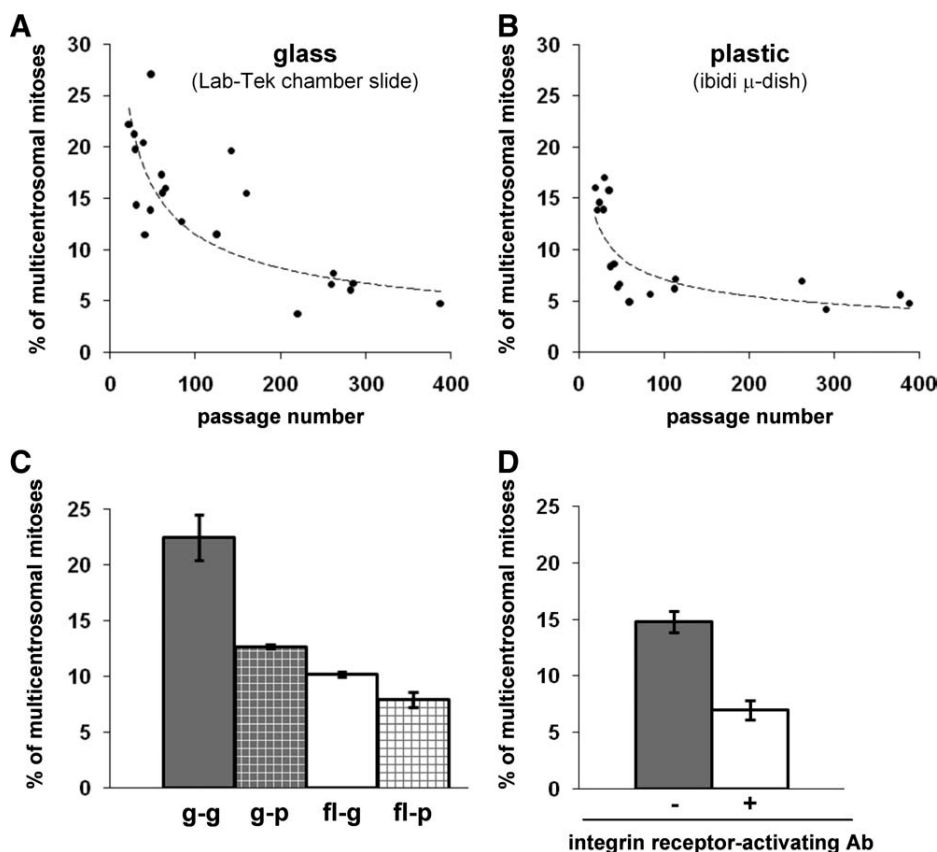


Figure 3. Both length of passaging and culture surface affect the frequency of multicentrosomal mitoses in human embryonic stem cells (hESCs). (A, B): Cells of the CCTL14 hESC line were propagated in culture for varying, prolonged periods and the frequency of multicentrosomal mitoses was analyzed. The frequency of multicentrosomal mitoses was determined in hESCs seeded on gelatinized glass (A) and gelatinized plastic (B). Each data point represents 60–300 mitoses evaluated. (C): The frequency of multicentrosomal mitoses in hESCs cultured on various substrates: gelatin-coated glass (g-g), gelatin-coated plastic (g-p), fibronectin/laminin-coated glass (fl-g), or fibronectin/laminin-coated plastic (fl-p). (D): Decreased frequency of multicentrosomal mitoses in undifferentiated hESCs treated for 24 hours with 10 μ g/ml integrin receptor-activating antibody. Data in (C) and (D) are expressed as the mean $\pm$ SEM of three independent replicates. Abbreviation: Ab, antibody.

to grow hESC colonies. As suboptimal culture conditions have been suggested to contribute to chromosomal instability in hESCs [23], we investigated the effect of interaction between hESCs and culture substratum on centrosome numbers in more detail. Normally, gelatin coating is sufficient for adhesion of hESCs onto the culture surface. Adhesion can be significantly improved by using a combination of fibronectin and laminin as the adhesive protein layer [24–26]. Thus, we have used these two conditions to provide qualitatively different substratums for hESC attachment. We have also introduced glass (chambers) or polyethylene-based (μ -dishes) cultureware, producing a total of four different combinations. As expected, the most frequent centrosomal abnormalities were observed in hESCs grown on glass coated with gelatin (average $\sim 22.5\%$), whereas the lowest frequency was seen in cells grown on polyethylene-based plastic coated with the fibronectin/laminin mix (average $\sim 8\%$). The remaining two conditions produced frequencies in between these extremes (Fig. 3C). It is of note that, in high-passage hESCs, an already low percentage of multicentrosomal mitoses remained unaffected by substratum type (not shown).

As fibronectin and laminin provide binding sites for integrins and propagate signals to cells through these receptors, we hypothesized that the beneficial effect of these extracellular matrix proteins was due to integrin-mediated signaling. We treated cultured hESCs with an integrin receptor-activating antibody for 24 hours to mimic the effect of stimulation by extracellular matrix proteins. Even such short-term artificial stimulation of integrins diminished the percentage of multicentrosomal mitoses (Fig. 3D).

Multicentrosomal Mitoses Occur in Both Hyperploid and Euploid hESCs

Supernumerary centrosomes are likely produced by either excessive rounds of duplication in one cell cycle or mitotic failure with subsequent tetraploidization associated with a cell having two centrosomes in G1 phase instead of one. We calculated that mitotic hESCs with four centrosomes accounted for about 60% of all multicentrosomal mitoses (supporting information Fig. S3). Although such numbers suggest that incomplete cell division (mitotic failure) underlies supernumerary centrosomes, a more rigorous analysis was needed to discriminate between the two above-mentioned possibilities. To this end, we simultaneously analyzed individual cells for centrosomal status using pericentrin staining and chromosome/chromatids number using kinetochore staining. Examples of kinetochores are shown in Figure 4A. Cells from three independent hESC lines at early and late passage were analyzed to obtain enough comprehensive data. Forty mitoses with supernumerary centrosomes were evaluated in each group. The most important finding was that the minimum ($\sim 28\%$, high-passage CCTL12 cells) and maximum ($\sim 73\%$, low-passage CCTL14 cells) mitoses with supernumerary centrosomes, depending on the cell line and passage number, were associated with a normal number of kinetochores ($23 \times 4 = 92$) so that the cells were euploid (Fig. 4B). The rest were then mitoses in which the number of kinetochores accounted for chromosomal multiplications. Although tetraploidies accounted for most of these multiplications, approximately 4%–10% of hESC kinetochore numbers far exceeded 8C (maximum $\sim 15C$). Taken together, the data from this analysis suggest that supernumerary centrosomes in hESCs are produced both by overduplication in one cell cycle and by mitotic failure.

To gain further support for this idea, we took advantage of the protein, Cep170, which resides only on the mother centriole during G1 phase and not on its nonmature daughter partner [27]. During G2 phase, cells normally contain two

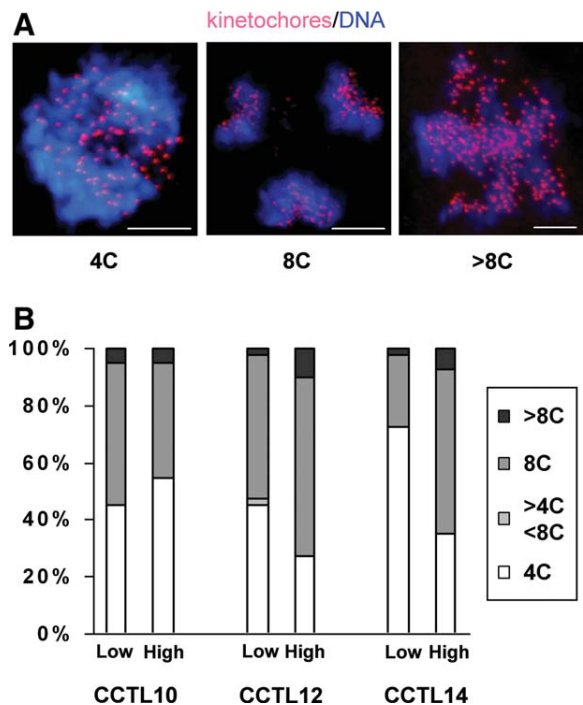


Figure 4. Multicentrosomal mitoses develop in both polyploid and diploid human embryonic stem cells (hESCs). (A): Representative images of mitotic hESCs in which the presence of supernumerary centrosomes is associated with different numbers of chromatids (4C, normal ploidy; $\geq 8C$, increased ploidy). Compiled stacks of confocal images are shown. Kinetochores were labeled with anti-centromere autoantibody (red), and chromatin was stained with 4,6'-diamidino-2-phenylindole (DAPI, blue). (B): Proportion of supernumerary centrosome-containing mitoses with normal (4C) and abnormal ($>4C$ to $<8C$, 8C, and $>8C$) numbers of chromatids. The numbers of kinetochores were determined by analyzing a series of confocal images of hESCs immunostained as described in (A). Only mitoses with supernumerary centrosomes were evaluated. Cells from three independent hESC lines of low (P16–P29) and high (P104–P167) passage were analyzed. Each column represents a total of 40 mitoses obtained in three independent replicates. Abbreviation: DAPI, 4,6'-diamidino-2-phenylindole.

centrosomes, each composed of one mature (Cep170-positive) and one nonmature (Cep170-negative) centriole. Our experiment was based on the rationale (summarized in supporting information Fig. S4) that the presence of any centrosome having more than one Cep170-negative centriole during G1 phase would show that reduplication took place in one cell cycle. The presence of two centrosomes in G1 phase, each having one Cep170-positive and one Cep170-negative centriole, would show that the extra centrosome was produced by mitotic failure. As we expected, based on our kinetochore countings, we observed both situations (supporting information Fig. S4C), further confirming the contribution of both mechanisms to numerical centrosomal abnormalities in hESCs.

Aurora A and CDK2 Both Contribute to Development of Supernumerary Centrosomes in hESCs

The activities of Aurora A and CDK2 have been linked to centrosomal metabolism. Overexpression of Aurora A in cells causes hyperploidy associated with hyperamplification of centrosomes [28]. Correspondingly, hyperamplification of chromosomes and centrosomes is common to many cancers, where it is often paralleled by overabundant Aurora

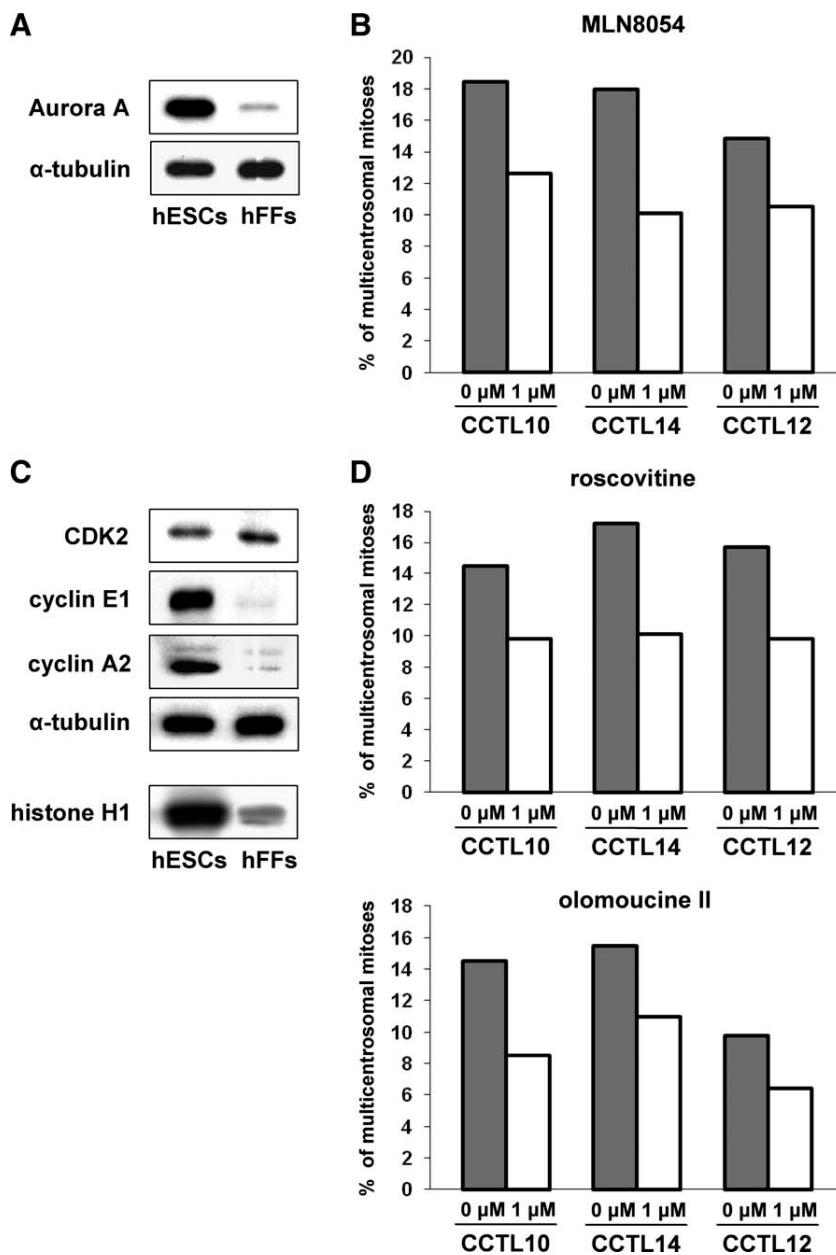


Figure 5. The key molecular regulators of centrosomal metabolism, CDK2 and Aurora A, participate in generation of supernumerary centrosomes in hESCs. (A): Aurora A protein levels are higher in hESCs than in hFFs. (B): Decreased frequency of multicentrosomal mitoses in undifferentiated hESCs treated with the selective Aurora A inhibitor, MLN8054. (C): Protein levels of the CDK2 activators, Cyclin E1 and A2, are higher in hESCs than in hFFs. CDK2 activity, as measured by in vitro kinase assay (with histone H1 as substrate), is also higher in hESCs. (D): Decreased frequency of multicentrosomal mitoses in undifferentiated hESCs treated with either of the CDK2 inhibitors, roscovitine or olomoucine II. Alpha-tubulin served as a loading control. Data in (B) and (D) represent three independent replicates for each inhibitor. Abbreviations: hESCs, human embryonic stem cells; hFFs, human foreskin fibroblasts.

A protein [29, 30]. Unrestrained activity of CDK2 has also been linked to overamplification of centrosomes; however, it does not affect cell ploidy [31, 32]. We found that Aurora A protein levels were much higher in growing hESCs than in human somatic cells (Fig. 5A). CDK2 activity was also very high in hESCs, probably due to high levels of activating cyclins E1 and A2 (Fig. 5C). Thus, both Aurora A and CDK2 are strong candidates for contributors to deregulation of centrosome numbers. Importantly, these factors act via two different mechanisms, that is, mitotic failure in the case of Aurora A and overamplification of centrosomes in one cell cycle in the case of CDK2. This duality was obviously in agreement with the results from our analysis of cell ploidy and supernumerary centrosomes. Therefore, we assessed the involvement of both Aurora A and CDK2 by analyzing the

effect of their pharmacological inhibition on the occurrence of multicentrosomal mitoses. To do so, we exposed growing hESCs to the Aurora A inhibitor, MLN8054, for 24 hours or either of the CDK2 inhibitors, roscovitine or olomoucine II, for 16 hours. As hypothesized, each individual treatment markedly reduced the frequency of multicentrosomal mitoses (Fig. 5B, 5D).

DISCUSSION

In the last decade, a large amount of published data has established an unequivocal link between centrosomal abnormalities and a cancerous phenotype [33]. For the first time, we have

linked abnormal centrosome numbers to a pristine hESC condition. Here, we show that, early after derivation from human blastocysts, these cells possess extra numbers of centrosomes during mitosis with an incidence ranging from 10% to 24%. We also show that more than one mechanism contributes to the appearance of supernumerary centrosomes in hESCs and that prolonged propagation of hESCs in culture leads to acquisition of unidentified molecular circuits, allowing these cells to eliminate/overcome causes of centrosomal overamplification.

As already mentioned, multicentrosomal mitoses are observed in hESCs when they (a) are in an undifferentiated state and (b) have not yet adapted to the unnatural environment of *in vitro* conditions. Although centrosomes are generally accepted as organelles that are mainly responsible for the faithful distribution of genetic material to daughter cells during mitosis, a role for centrosomes in other cellular processes is emerging. The most relevant in this context is the importance of the asymmetric behavior of centrosomes in the regulation of neural and germ stem cell fates [34–37]. Although these studies are not directly linked to hESCs, they still point to the existence of centrosome-mediated regulatory pathways, which are unique to stem/progenitor cells and are absent from somatic cells. If a certain uniqueness of centrosomal metabolism is inherent to inner cell mass (ICM) cells and also to the hESCs originating from them, one can understand that only pristine hESCs produce supernumerary centrosomes and are unexpectedly tolerant to them. Such a notion is consistent with the finding by Mantel et al. [38] that undifferentiated hESCs are exquisitely tolerant to polyploidy and do not die under unfavorable genetic conditions, thanks to the uncoupling of the spindle checkpoint from the cell death pathway. It is of note that the switch from “tolerant” to “nontolerant” molecular circuitries may account for our unpublished observations in the CCTL10 line. When grown under undifferentiating conditions, these cells typically have a very high incidence of supernumerary centrosomes; however, immediately on initiation of differentiation by prevention of substratum attachment, massive cell death takes place and no EBs are formed. Our data on centrosomes, combined with the findings reported by Mantel et al. [38], support the speculative hypothesis that, very early in embryo development, cell quantity is preferred over quality and surveillance mechanisms are fully activated only after sufficient cell mass is generated.

Centrosomal amplification and structural centrosomal abnormalities are rare in most non-transformed somatic cell types. The centrosomal duplication cycle seems to be supervised by p53, which inhibits CDK activity through p21 [39]. Inactivation of p53 increases susceptibility to abnormal amplification of centrosomes [40]. In hESCs, the classical p53-p21 axis does not link cell cycle-driving CDKs and ultraviolet radiation-C range (UVC) light-induced cellular stress due to the lack of accumulation of p21 protein [41]. Obviously, the p53-mediated mechanism preventing overamplification of centrosomes in somatic cells is inoperative in hESCs, as the cells of the CCTL14 hESC line used here carry nonmutated p53 (determined by functional yeast-based assay, data not shown) and are also capable of accumulating p53 [41].

Given the findings discussed earlier, one might assume that the configuration of surveillance pathways, which is unique to ICM-born cells, is the sole source of supernumerary centrosomes in hESCs. However, our results also point to at least three other molecular elements that contribute to the high incidence of numerical centrosomal abnormalities in hESCs.

First, we have found that improving quality of interaction between hESCs and culture substratum counteracts overamplification of centrosomes and that this phenomenon is most

likely mediated by an increased activation of integrins. Our finding is not the first indication of a functional link between molecules of the adhesion complex and centrosomal metabolism, as Pugacheva et al. have suggested that such interplay exists in non-ESC types [42].

Second, we have found that Aurora A, the mitotic kinase with oncogenic potential [43, 44], was highly expressed in hESCs and inhibition of its activity reduced the frequency of multicentrosomal mitoses in these cells. Indeed, several recent studies have shown that Aurora A is involved in the regulation of progression through G2/M phase in non-ESCs [45, 46]. Therefore, we hypothesize that the reduced frequency of multicentrosomal mitoses after Aurora A inhibition is caused by delayed mitosis entry. According to this scenario, cells with normal centrosomes do not require high levels of Aurora A and smoothly proceed to mitosis, even when this kinase is chemically inhibited, whereas this is not the case with cells harboring centrosomal abnormalities. As a result, the mitotic cell population is relatively enriched for normal two-centrosomal mitoses, as we have seen in our analysis.

Third, we have shown that suppressing CDK2 activity in hESCs decreases the frequency of mitoses with supernumerary centrosomes. Normally, centrosome duplication and DNA replication are synchronized, with CDK2 being responsible for concurrent initiation of both processes [47–49]. This coordination may be abrogated in situations when CDK2 activity is persistently high. Under such conditions, centrosome and DNA cycles may become uncoupled, and multiple cycles of centrosome duplication may take place [50, 51]. We propose that such dysregulation occurs, at least to some extent, in pluripotent cells, which exhibit high and probably less fluctuating CDK2 activity [52].

In principle, supernumerary centrosomes/MTOCs may arise through (a) excessive rounds of centrosome duplication in a single cell cycle (centrosome overduplication), (b) cell division failure resulting in doubling of both the genome and centrosomes, (c) fragmentation of centrosome, and (d) *de novo* formation of acentriolar PCM into foci functioning as MTOCs. The latter two possibilities are very unlikely to take place in hESCs, as we have failed to detect with any structural abnormalities using confocal or TEM. In contrast, the first two mechanisms clearly contribute to generation of supernumerary centrosomes in hESCs as seen by the association of multicentrosomal mitotic states with both normal and increased ploidies ($\geq 4C$; see Fig. 4). Moreover, we cannot exclude the possibility that these two mechanisms operate in combined manner, as depicted in Figure 6. This notion is supported by the presence of Cep170-containing centrioles in combinations that correspond to both mechanisms (supporting information Fig. S4). Notably, the proportion of cells having supernumerary centrosomes due to differing mechanisms is strongly affected by both hESC line and passage number. This underscores the complexity of cues that influence centrosomal metabolism in hESCs. It may also help explain the differences in predisposition to karyotype instability observed between hESC lines [53].

Although hESCs tend to accumulate genetic changes during their propagation in culture, they are considered to be non-transformed, at least at the time of their derivation and at early passages. Here, we demonstrate, on the large panel of independent hESC lines, that early passage hESCs, but not hESCs that have been propagated long-term, accumulate supernumerary centrosomes at a high frequency. Prolonged propagation of hESCs in culture is associated with a continuous decline in the frequency of mitoses with supernumerary centrosomes, with these frequencies ultimately reaching the levels observed in somatic cells such as hFFs (data shown here), human dermal

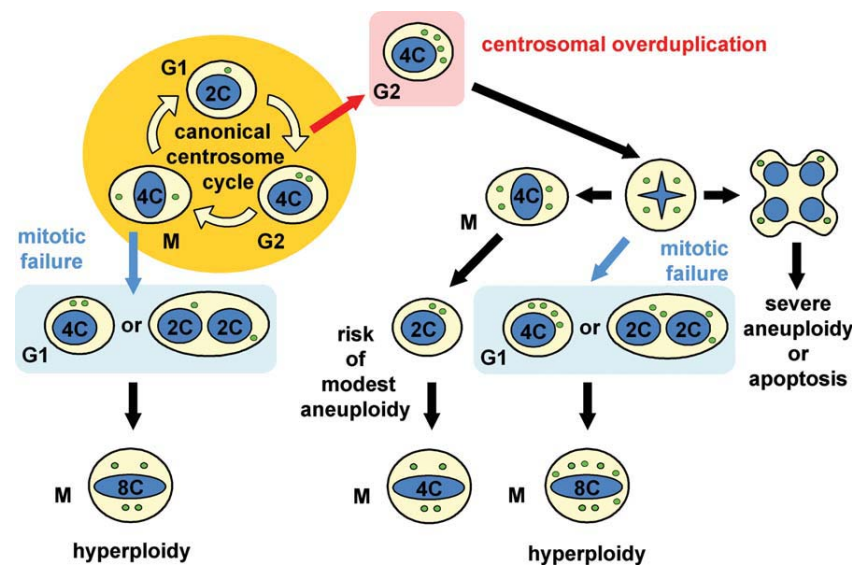


Figure 6. Schematic showing mechanisms underlying supernumerary centrosomes in human embryonic stem cells and possible outcomes. Normally (yellow area), a cell enters mitosis with replicated DNA (gene copy number, C; four chromatids, 4C) and two copies of the centrosome. Supernumerary centrosomes may be produced by either mitotic failure (blue arrow, blue rectangle) or excessive rounds of centrosomal duplication during interphase (red arrow, pink rectangle). In mitotic failure, a normal bipolar spindle is assembled, but cell division is not completed, producing a mononuclear or binuclear tetraploid (4C) cell with two centrosomes instead of one. The tetraploidy and extra centrosomes may be tolerated, and the cell may continue cycling, causing its chromosomal and centrosomal complement to double. Supernumerary centrosomes produced by overduplication (pink rectangle) may lead to three possible outcomes (shown from the left to the right). First, supernumerary centrosomes will become clustered on bipolar axis forming a “pseudobipolar spindle” that structurally resembles a normal bipolar division spindle. This strategy is not void of risk of segregation errors and may produce a modest degree of aneuploidy that does not compromise cellular ability to survive. Second, a multipolar spindle will be assembled and the mitotic failure will occur. Again, tetraploidy and extra centrosomes may be tolerated, and the cell may continue cycling. Third, a multipolar spindle will be assembled, and cell division will proceed, generating progeny that have severe aneuploidy and is likely to die. Both mitotic failure and multiple rounds of centrosomal duplication may occur in combination, making the centrosomal/chromosomal configurations even more complex.

fibroblasts [54], CD34+ umbilical cord blood cells [55], ovarian surface epithelial cells [56], and oral mucosa [57]. One can hypothesize that supernumerary centrosomes are among the targets of the selection pressure encountered by hESCs during adaptation to *in vitro* conditions. Regardless of the nature of changes (genetic or epigenetic) that develop in hESCs to counteract sources of supernumerary centrosomes, the way that these changes spread in the hESC population most likely follows a scenario known as the “genetic convergence model” [58, 39]. According to this model, which is thought to apply to at least some cancers, supernumerary centrosomes and genetic heterogeneity diminish after certain time, suggesting that the clone with the genetic configuration optimal for growth in given environment dominates in culture.

Our experiments clearly show that pristine hESCs possess a precondition, in the form of supernumerary centrosomes, for development of chromosomal abnormalities. However, whether this precondition really represents a major risk to hESCs in terms of their chromosomal stability remains unclear. One argument against this idea is that finalized multipolar cytokinesis is a rare event and such division may generate only severely hypoploid progeny with grossly compromised viability. Still, cells can also coalesce (cluster) supernumerary centrosomes into just two large aggregates that act as MTOCs of the dominant spindle and then proceed through mitosis with such pseudobipolar spindles [59]. Importantly, a series of studies have recently demonstrated that centrosomal clustering does not necessarily faithfully resolve the burden of multipolar prometaphases, as even only a transitional presence of multipolar spindles in early mitosis facilitates establishment of kinetochore misattachments [14, 15]. Residual merotelic attachments then produce, in high fre-

quency, individual chromosome missegregation due to lagging of chromosomes in anaphase. Although, we have noticed lagging chromosomes and unequal pseudobipolar cell divisions (supporting information Fig. S5) in our hESC preparations, whether they result from merotely is currently being investigated by monitoring multicentrosomal mitoses in live cells.

Finally, it must be noted that the conventional model of centrosomal dysfunction leading to genomic instability may not necessarily account for all possible negative effects of abnormal centrosomal metabolism. For example, positioning of centrosomes has been recently shown to govern symmetric versus asymmetric cell division and to be causally linked to cell transformation [60, 61]. In addition, there are myriads of centrosome-associated proteins with mostly unexplored functions, so, one can hypothesize that the perturbed distribution of these factors due to abnormal centrosome number may affect pluripotency-related processes in hESCs. Therefore, these and other hypotheses need further testing.

CONCLUSION

Undifferentiated hESCs suffer from a high frequency of multicentrosomal mitoses. These supernumerary centrosomes are a product of a combination of several factors including less proficient surveillance/checkpoint pathways, high activity of CDK2 and Aurora A, and unfavorable culture conditions. Although the data are not conclusive on whether supernumerary centrosomes contribute to chromosomal instability in hESCs, they identify supernumerary centrosomes as a

significant risk factor and most importantly, show that there are strategies available to reduce this risk.

ACKNOWLEDGMENTS

We thank Dr. Outi Hovatta (Karolinska Institutet, Stockholm, Sweden) for samples of cultured hESCs, Dr. Spiros Linardopoulos (Millenium Pharmaceuticals, Cambridge, MA) for Aurora A inhibitor, MLN8054, Dr. Miroslav Strnad (Palacký University Olomouc, Olomouc, Czech Republic) for olomoucine II, Dr. Mark Ginsberg (University of California San Diego, CA) for anti-LIBS-6 $\alpha_{IIb}\beta_3$ integrin receptor-activating antibody, and Dr. Pavel Dráber (Institute of Molecular Genetics, Academy of Sciences of the Czech Republic, Prague, Czech Republic) for anti-

γ -tubulin antibody (TU-31). We are grateful to Iveta Peterková, Dr. Martina Vodinská, and Dr. Klára Koudelková for excellent technical assistance. This work was supported by the Czech Science Foundation (204/09/2044), Ministry of Education, Youth, and Sports of the Czech Republic (MSM0021622430, 1M0538, 2B06052), the Academy of Sciences of the Czech Republic (AV0Z50390512, AV0Z50390703), and EU (LSHG-CT-2006-018739).

DISCLOSURE OF POTENTIAL CONFLICTS OF INTEREST

The authors indicate no potential conflicts of interest.

REFERENCES

- Blum B, Benvenisty N. The tumorigenicity of diploid and aneuploid human pluripotent stem cells. *Cell Cycle* 2009;8:3822–3830.
- Draper JS, Smith K, Gokhale P et al. Recurrent gain of chromosomes 17q and 12 in cultured human embryonic stem cells. *Nat Biotechnol* 2004;22:53–54.
- Maitra A, Arking DE, Shivapurkar N et al. Genomic alterations in cultured human embryonic stem cells. *Nat Genet* 2005;37:1099–1103.
- Mitalipova MM, Rao RR, Hoyer DM et al. Preserving the genetic integrity of human embryonic stem cells. *Nat Biotechnol* 2005;23:19–20.
- Schvartzman J, Sotillo R, Benezra R. Mitotic chromosomal instability and cancer: Mouse modelling of the human disease. *Nat Rev Cancer* 2010;10:102–115.
- D'Assoro AB, Barrett SL, Folk C et al. Amplified centrosomes in breast cancer: A potential indicator of tumor aggressiveness. *Breast Cancer Res Treat* 2002;75:25–34.
- Pihan GA, Wallace J, Zhou Y et al. Centrosome abnormalities and chromosome instability occur together in pre-invasive carcinomas. *Cancer Res* 2003;63:1398–1404.
- Chng WJ, Braggio E, Mulligan G et al. The centrosome index is a powerful prognostic marker in myeloma and identifies a cohort of patients that might benefit from aurora kinase inhibition. *Blood* 2008;111:1603–1609.
- Alieva IB, Uzbekov RE. The centrosome is a polyfunctional multiprotein cell complex. *Biochemistry (Mosc)* 2008;73:626–643.
- Meraldi P, Nigg EA. The centrosome cycle. *FEBS Lett* 2002;521:9–13.
- Nigg EA. Centrosome duplication: Of rules and licenses. *Trends Cell Biol* 2007;17:215–221.
- Rusan NM, Rogers GC. Centrosome function: Sometimes less is more. *Traffic* 2009;10:472–481.
- Schatten H. The mammalian centrosome and its functional significance. *Histochem Cell Biol* 2008;129:667–686.
- Ganem NJ, Godinho SA, Pellman D. A mechanism linking extra centrosomes to chromosomal instability. *Nature* 2009;460:278–282.
- Silkworth WT, Nardi IK, Scholl LM et al. Multipolar spindle pole coalescence is a major source of kinetochore mis-attachment and chromosome mis-segregation in cancer cells. *Plos One* 2009;4:e6564.
- Adewumi O, Aflatoonian B, Ahrlund-Richter L et al. Characterization of human embryonic stem cell lines by the International Stem Cell Initiative. *Nat Biotechnol* 2007;25:803–816.
- Manfredi MG, Ecsedy JA, Meetze KA et al. Antitumor activity of MLN8054, an orally active small-molecule inhibitor of Aurora A kinase. *Proc Natl Acad Sci USA* 2007;104:4106–4111.
- Krystof V, McNae IW, Walkinshaw MD et al. Antiproliferative activity of olomoucine II, a novel 2,6,9-trisubstituted purine cyclin-dependent kinase inhibitor. *Cell Mol Life Sci* 2005;62:1763–1771.
- Frelinger AL, Du XP, Plow EF et al. Monoclonal antibodies to ligand-occupied conformers of integrin $\alpha_{IIb}\beta_3$ (glycoprotein IIb-IIIa) alter receptor affinity, specificity, and function. *J Biol Chem* 1991;266:17106–17111.
- Reverte CG, Benware A, Jones CW et al. Perturbing integrin function inhibits microtubule growth from centrosomes, spindle assembly, and cytokinesis. *J Cell Biol* 2006;174:491–497.
- Matula P, Maška M, Danek O et al. Software for the Acquisition and Analysis of 3D Images of Cells in Fluorescence Microscopy: Proceedings of the IEEE International Symposium on Biomedical Imaging, Bos-

- ton, IEEE New York. 2009:1138–1141. Available at <http://cbia.fi.muni.cz/acquarium/>.
- Lingle WL, Salisbury JL. Altered centrosome structure is associated with abnormal mitoses in human breast tumors. *Am J Pathol* 1999;155:1941–1951.
- Baker DEC, Harrison NJ, Maltby E et al. Adaptation to culture of human embryonic stem cells and oncogenesis in vivo. *Nat Biotechnol* 2007;25:207–215.
- Amit M, Shariki C, Margulets V et al. Feeder layer- and serum-free culture of human embryonic stem cells. *Biol Reprod* 2004;70:837–845.
- Draper JS, Moore HD, Ruban LN et al. Culture and characterization of human embryonic stem cells. *Stem Cells Dev* 2004;13:325–336.
- Xu C, Inokuma MS, Denham J et al. Feeder-free growth of undifferentiated human embryonic stem cells. *Nat Biotechnol* 2001;19:971–974.
- Guarguaglini G, Duncan PI, Stierhof YD et al. The forkhead-associated domain protein Cep170 interacts with Polo-like kinase 1 and serves as a marker for mature centrioles. *Mol Biol Cell* 2005;16:1095–1107.
- Meraldi P, Honda R, Nigg EA. Aurora-A overexpression reveals tetraploidization as a major route to centrosome amplification in p53^{-/-} cells. *EMBO J* 2002;21:483–492.
- Katayama H, Brinkley WR, Sen S. The Aurora kinases: Role in cell transformation and tumorigenesis. *Cancer Metastasis Rev* 2003;22:451–464.
- Yamamoto Y, Eguchi S, Junpei A et al. Intercellular centrosome number is correlated with the copy number of chromosomes in bladder cancer. *Cancer Genet Cytogenet* 2009;191:38–42.
- Lingle WL, Lukasiewicz K, Salisbury JL. Deregulation of the centrosome cycle and the origin of chromosomal instability in cancer. *Adv Exp Med Biol* 2005;570:393–421.
- Fukasawa K. Centrosome amplification, chromosome instability and cancer development. *Cancer Lett* 2005;230:6–19.
- Zyss D, Gergely F. Centrosome function in cancer: Guilty or innocent? *Trends Cell Biol* 2009;19:334–346.
- Rebollo E, Sampaio P, Januschke J et al. Functionally unequal centrosomes drive spindle orientation in asymmetrically dividing Drosophila neural stem cells. *Dev Cell* 2007;12:467–474.
- Rusan NM, Peifer M. A role for a novel centrosome cycle in asymmetric cell division. *J Cell Biol* 2007;177:13–20.
- Yamashita YM, Mahowald AP, Perlin JR et al. Asymmetric inheritance of mother versus daughter centrosome in stem cell division. *Science* 2007;315:518–521.
- Wang X, Tsai J, Imai JH et al. Asymmetric centrosome inheritance maintains neural progenitors in the neocortex. *Nature* 2009;461:947–955.
- Mantel C, Guo Y, Lee MR et al. Checkpoint-apoptosis uncoupling in human and mouse embryonic stem cells: A source of karyotypic instability. *Blood* 2007;109:4518–4527.
- Fukasawa K. P53, cyclin-dependent kinase and abnormal amplification of centrosomes. *Biochim Biophys Acta* 2008;1786:15–23.
- Fukasawa K, Choi T, Kuriyama R et al. Abnormal centrosome amplification in the absence of p53. *Science* 1996;271:1744–1747.
- Bárta T, Vinarský V, Holubcová Z et al. Human embryonic stem cells are capable of executing G1/S checkpoint activation. *Stem Cells* 2010;28:1143–1152.
- Pugacheva EN, Roegiers F, Golemis EA. Interdependence of cell attachment and cell cycle signaling. *Curr Opin Cell Biol* 2006;18:507–515.
- Zhou H, Kuang J, Zhong L et al. Tumour amplified kinase STK15/BTAK induces centrosome amplification, aneuploidy and transformation. *Nat Genet* 1998;20:189–193.

- 44 Bischoff JR, Anderson L, Zhu Y et al. A homologue of *Drosophila* aurora kinase is oncogenic and amplified in human colorectal cancers. *EMBO J* 1998;17:3052–3065.
- 45 Liu Q, Ruderman JV. Aurora A, mitotic entry, and spindle bipolarity. *Proc Natl Acad Sci USA* 2006;103:5811–5816.
- 46 Sasayama T, Marumoto T, Kunitoku N et al. Over-expression of Aurora-A targets cytoplasmic polyadenylation element binding protein and promotes mRNA polyadenylation of Cdk1 and cyclin B1. *Genes Cells* 2005;10:627–638.
- 47 Hinchcliffe EH, Li C, Thompson EA et al. Requirement of Cdk2-cyclin E activity for repeated centrosome reproduction in *Xenopus* egg extracts. *Science* 1999;283:851–854.
- 48 Meraldi P, Lukas J, Fry AM et al. Centrosome duplication in mammalian somatic cells requires E2F and Cdk2-cyclin A. *Nat Cell Biol* 1999;1:88–93.
- 49 Lacey KR, Jackson PK, Stearns T. Cyclin-dependent kinase control of centrosome duplication. *Proc Natl Acad Sci USA* 1999;96:2817–2822.
- 50 Balczon R, Bao L, Zimmer WE et al. Dissociation of centrosome replication events from cycles of DNA synthesis and mitotic division in hydroxyurea-arrested Chinese hamster ovary cells. *J Cell Biol* 1995;130:105–115.
- 51 Mussman JG, Horn HF, Carroll PE et al. Synergistic induction of centrosome hyperamplification by loss of p53 and cyclin E overexpression. *Oncogene* 2000;19:1635–1646.
- 52 Stead E, White J, Faast R et al. Pluripotent cell division cycles are driven by ectopic Cdk2, cyclin A/E and E2F activities. *Oncogene* 2002;21:8320–8333.
- 53 Catalina P, Montes R, Ligerio G et al. Human ESCs predisposition to karyotypic instability: Is a matter of culture adaptation or differential vulnerability among hESC lines due to inherent properties? *Mol Cancer* 2008;7:76–84.
- 54 Fabarius A, Giehl M, Frank O et al. Induction of centrosome and chromosome aberrations by imatinib in vitro. *Leukemia* 2005;19:1573–1578.
- 55 Giehl M, Fabarius A, Frank O et al. Centrosome aberrations in chronic myeloid leukemia correlate with stage of disease and chromosomal instability. *Leukemia* 2005;19:1192–1197.
- 56 Hsu L, Kapali M, DeLoia JA et al. Centrosome abnormalities in ovarian cancer. *Int J Cancer* 2005;113:746–751.
- 57 Giehl M, Leitner A, Haferlach C et al. Detection of centrosome aberrations in disease-unrelated cells from patients with tumor treated with tyrosine kinase inhibitors. *Eur J Haematol* 2010;85:139–148.
- 58 Heim S, Mandahl N, Mitelman F. Genetic convergence and divergence in tumor progression. *Cancer Res* 1988;48:5911–5916.
- 59 Quintyne NJ, Reing JE, Hoffelder DR et al. Spindle multipolarity is prevented by centrosomal clustering. *Science* 2005;307:127–129.
- 60 Basto R, Brunk K, Vinadogrova T et al. Centrosome amplification can initiate tumorigenesis in flies. *Cell* 2008;133:1032–1042.
- 61 Quyn AJ, Appleton PL, Carey FA et al. Spindle orientation bias in gut epithelial stem cell compartments is lost in precancerous tissue. *Cell Stem Cell* 2010;6:175–181.



See www.StemCells.com for supporting information available online.